

can choose to put SSDs or hard drives inside your NAS. SSDs help with data caching while mechanical hard drives in RAID configuration will provide good transfer speeds.

What do the numbers mean?

Just like graphics cards, even storage benchmarks are an easy thing to go through. You'll never be able to benchmark storage media before buying them but the marketing material on the packaging should give you some idea of what you're going in for. Let's clear those for you.

Throughput

The storage media's absolute maximum data read/write speeds is what throughput tells you. A lot depends on the type of NAND and the controller if we're looking at storage media that uses flash memory. This is why some USB 3.0 drives outperform others by a wide margin though both are of the same capacity.

IOPS

IOPS has been a standard metric for storage devices since their existence. It stands for input/output operations per second. Every time the computer tries to access a sliver of data from the storage device, one read/output operation is performed. The same goes for whenever the computer tries to store data on the media. The input/output operations use data of different sizes which is why throughput isn't preferred and IOPS makes a better case. Depending on the size of each operation you can calculate the throughput using simple multiplication, $\text{Throughput} = \text{IOPS} \times \text{size}$. If your values are off then probably the operation involves data of different sizes or you've hit a bottleneck somewhere.

HOW TO TEST MORE

In order to maximise the capabilities of storage media, you need to run these tests from devices that run much faster, which is why we have SSDs in RAID 0 for our storage media tests. Each drive is prepped for our test sequence which requires the entire drive to be in a RAW state for certain tests while for other tests, the media has to be initialised to accept data. The drive to be tested is connected to an interface port that is natively linked to the CPU rather than via a third party chipset. This reduces latency between the media and the CPU. Also, third party chips have been found to be not at par with native ports in terms of throughput.

We use a plethora of software, some of which run the same tests, to gather as many numbers as we can. These tests are also run over and over again to generate multiple readings which are then averaged for the final score. Not all storage media require the same type of data sets. For example, if benchmarking an SD card you wouldn't be looking at throughput registered with file sizes that are 1GB or more. It is likely that you'll be using the SD card in a camera which might take images that are 12-40MB in size. So these are the numbers you're interested in. A queue depth of 4-5 will tell you about the performance of the drive when you are taking burst shots with your DSLR. We use ATTO and CrystalDiskMark primarily for flash based media.

Hard drives and SSDs are more complex, they require different tests since they handle files of all sizes at varying queue depths. Benchmarks with smaller file sizes indicate performance related to the operating system, while larger file sizes indicate your day-to-day copy/delete operations. In such situations, a trace based tool is used. PC Mark's storage benchmark and IOMeter allow you to replicate these operations. Hard drive operations are monitored in different usage scenarios and those operations are replicated by these software to produce a much accurate representation of their real world performance.

Queue Depth

Queue depth is pretty self-explanatory. When you have multiple operations lined up after one another you get a queue of operations. The number of operations lined up translates to the depth of this transfer queue. Now in hard drives, queue depth can include read and write operations and with AHCI (most newer drives will be configured to use this) the drive intelligently sorts the queue so that the head (the actual component that does the job of read/write) doesn't have to travel far for the next operation. This optimisation helps with the transfer speed.

On SSDs, this is slightly different since there is no physical head, seek times are next to nothing and to actually saturate an SSD's throughput rate you have to have an

excessively high queue depth. Needless to say, we used the term excessive since your day-to-day operations will never achieve such high queue depth.

Real World transfer speeds

This is that one metric that you need to look out for. Synthetic tests will maximise throughput on most drives but real world tests will not. It shows you how efficiently a storage device performs at lower queue depths which are going to form the bulk of its operations. These tests are performed using files of different sizes and the data may be uniform or random.

Seek times

This is a metric that is becoming less relevant as SSDs become more affordable. Seek time refers to the time taken by the mechanical head to get from its resting point to where the data is actually stored. SSDs operate in a different manner altogether and don't suffer from latency issues. There are a lot of eco-powered drives on the market which reduce the power provided to the hard drive when not in use. This drastically increases seek times which is why such drives aren't recommended as your primary OS drive. 



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